If an object of mass $m > 0$ is attached to a wall by a spring of stiffness $k > 0$ and slides around on a frictionless surface, then a combination of Newton's Law and Hooke's Law says that the displacement x of the object is governed by the ODE $mx'' = -kx$.

1. If $x(0) = 1$, which of the following could be $x(t)$?

(A) $x(t) = \cos(t\sqrt{k/m})$

(B) $x(t) = e^{t\sqrt{k/m}}$

(C) $x(t) = e^{-t\sqrt{k/m}}$

(D) None of the above

2. True or False?

If λ is a repeated eigenvalue of a 2×2 matrix, it must be defective.

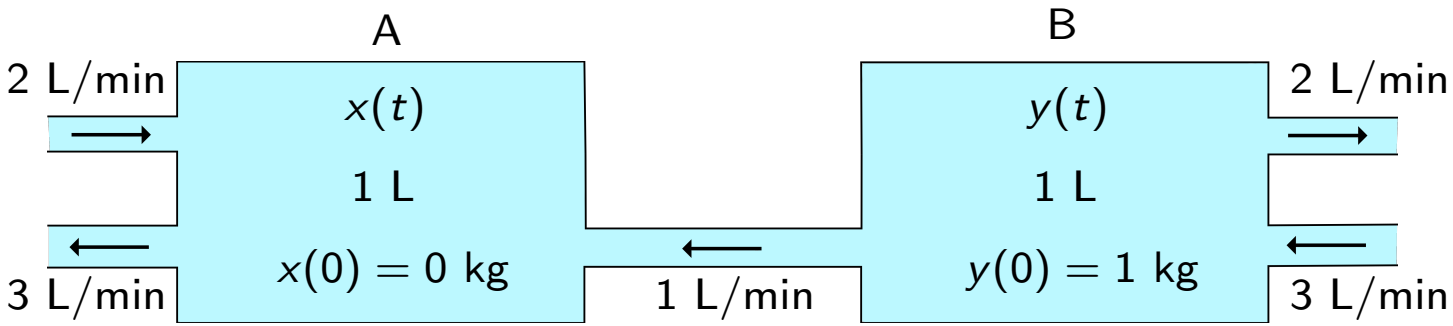
3. True or False?

If λ is a repeated eigenvalue of a 2×2 matrix of the form

$$A = \begin{bmatrix} 0 & 1 \\ b & a \end{bmatrix},$$

it must be defective.

More salt-water tanks! Hooray! In the diagram below, the inputs to both tanks A and B are pure water.



4. If A is the matrix such that $\dfrac{d}{dt}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = A \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$, which of the following is true?

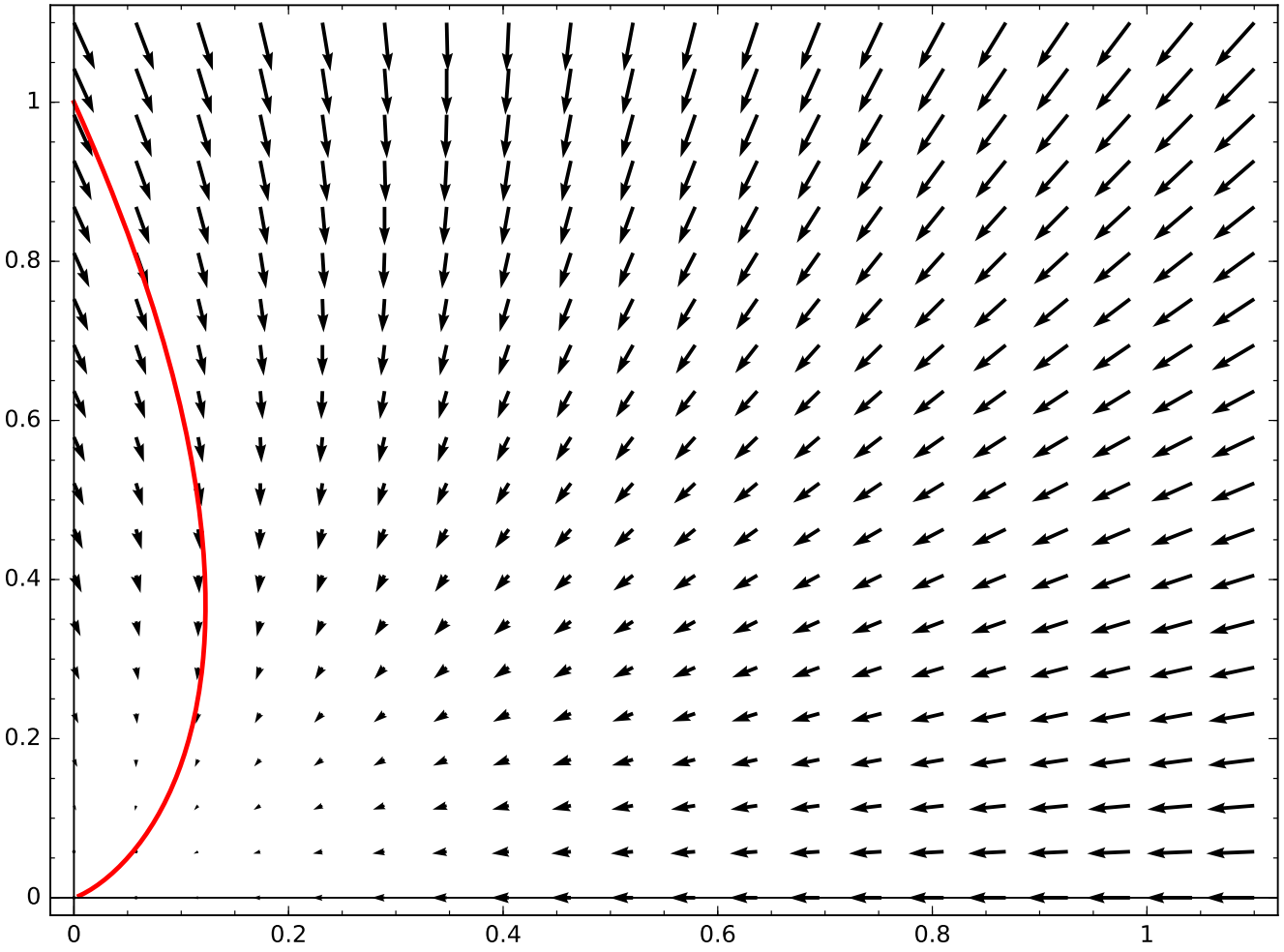
(A) A has a repeated deficient eigenvalue.

(B) A has a repeated complete eigenvalue.

(C) A has two distinct real eigenvalues.

(D) A has two distinct complex eigenvalues.

Here's a phase portrait for the previous problem.



If an object of mass $m = 1$ kg is attached to a wall by a spring of stiffness $k = 20$ kg/s^2 and slides around on a surface with coefficient of friction $b = 4$ kg/s, the displacement x of the object from its equilibrium position is described by the ODE

$$x'' + 4x' + 20x = 0.$$

5. True or False?

$\lim_{t \to \infty} x(t) = 0.$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

The characteristic polynomial of A is $-(\lambda - 2)^3$. Two linearly independent eigenvectors corresponding to the eigenvalue $\lambda = 2$ are $\vec{v}_1 = (1, 0, 1)$ and $\vec{v}_2 = (0, 1, 0)$.

6. Which of the following statements is true?

(A) There exists a third eigenvector $\vec{v}_3$ corresponding to the eigenvalue $\lambda = 2$.

(B) There exists a generalized eigenvector $\vec{v}_3$ such that $(A - 2I)\vec{v}_3 = \vec{v}_1$.

(C) There exists a generalized eigenvector $\vec{v}_3$ such that $(A - 2I)\vec{v}_3 = \vec{v}_2$.

(D) None of the above.

7. True or False?

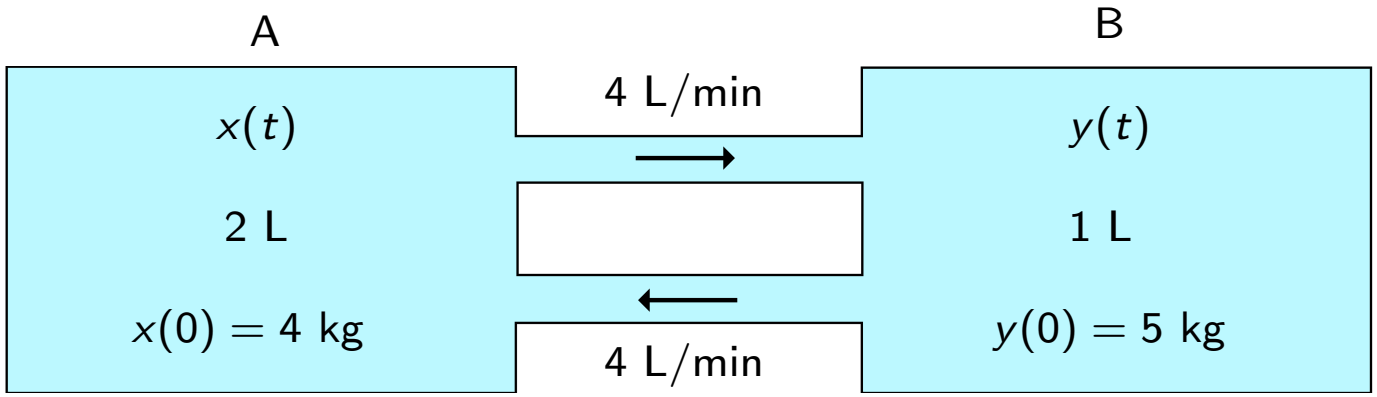
The phase portrait of the system $\vec{x}' = \begin{bmatrix} 1 & a \\ a & 1 \end{bmatrix} \vec{x}$ can never be a sink.

7. True or False?

The phase portrait of the system $\vec{x}' = \begin{bmatrix} 1 & a \\ a & 1 \end{bmatrix} \vec{x}$ can never be a sink.

Follow-up. What does the phase portrait look like for various possiblities of a?

More interconnected salt-water tanks!



8. True or False?

$$\lim_{t\to\infty} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

9. If we rewrite the linear system of ODEs

$$\begin{cases} x'' = t^2 x + y' + x' \\ y''' = tx' + \sin(t) y \end{cases}$$

as a first order system $\vec{x}\,' = A\vec{x}$ for a matrix A, what are the dimensions of the matrix A?

(A) 2×2

(B) 3×3

(C) 4×4

(D) 5×5

10. Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Which of the following vectors is a generalized eigenvector of A but *not* an eigenvector?

(A) $\vec{e}_1 = (1, 0, 0)$

(B) $\vec{e}_2 = (0, 1, 0)$

(C) $\vec{e}_3 = (0, 0, 1)$

(D) None of the above

Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$.

Observe that $\lambda = 2$ is the only eigenvalue, that $\vec{e}_1$ is an eigenvector, that $(A - 2I)\vec{e}_2 = \vec{e}_1$ and that $(A - 2I)\vec{e}_3 = \vec{e}_2$.

11. Which of the following is *not* a solution of the system $\vec{x}\,' = A\vec{x}$?

(A) $\vec{x}(t) = e^{2t}(1, 0, 0)$

(B) $\vec{x}(t) = e^{2t}(t, 1, 0)$

(C) $\vec{x}(t) = e^{2t}(0, t, 1)$

(D) $\vec{x}(t) = e^{2t}(t^2/2, t, 1)$